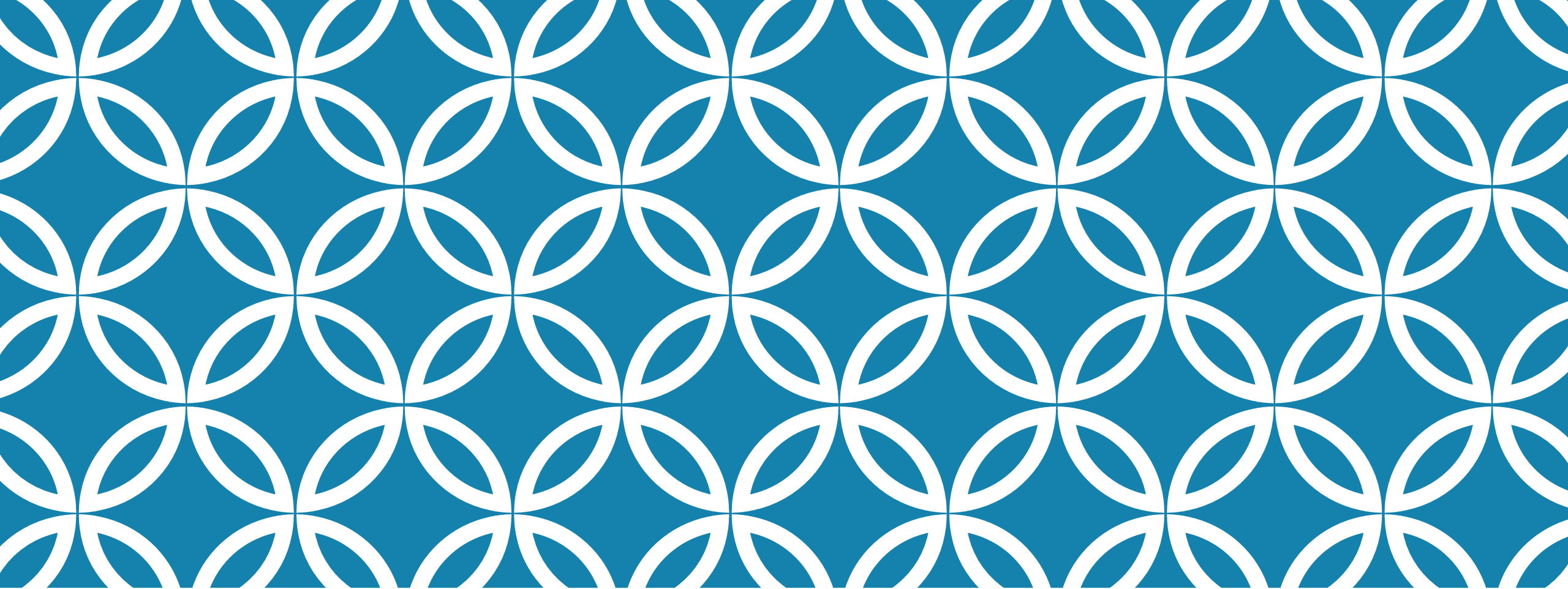




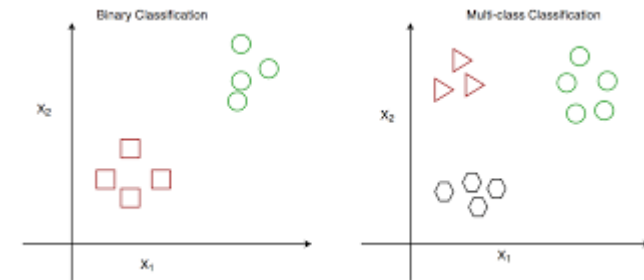
LOGISTIC REGRESSION

MACHINE LEARNING
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VIT Chennai

MODELLING CLASS PROBABILITIES VIA LOGISTIC REGRESSION

Logistic regression is a model for **classification task**, not for **regression task**.

Despite the terminology says regression, it is not a regression model at all. Rather, it is a **binary classifier**

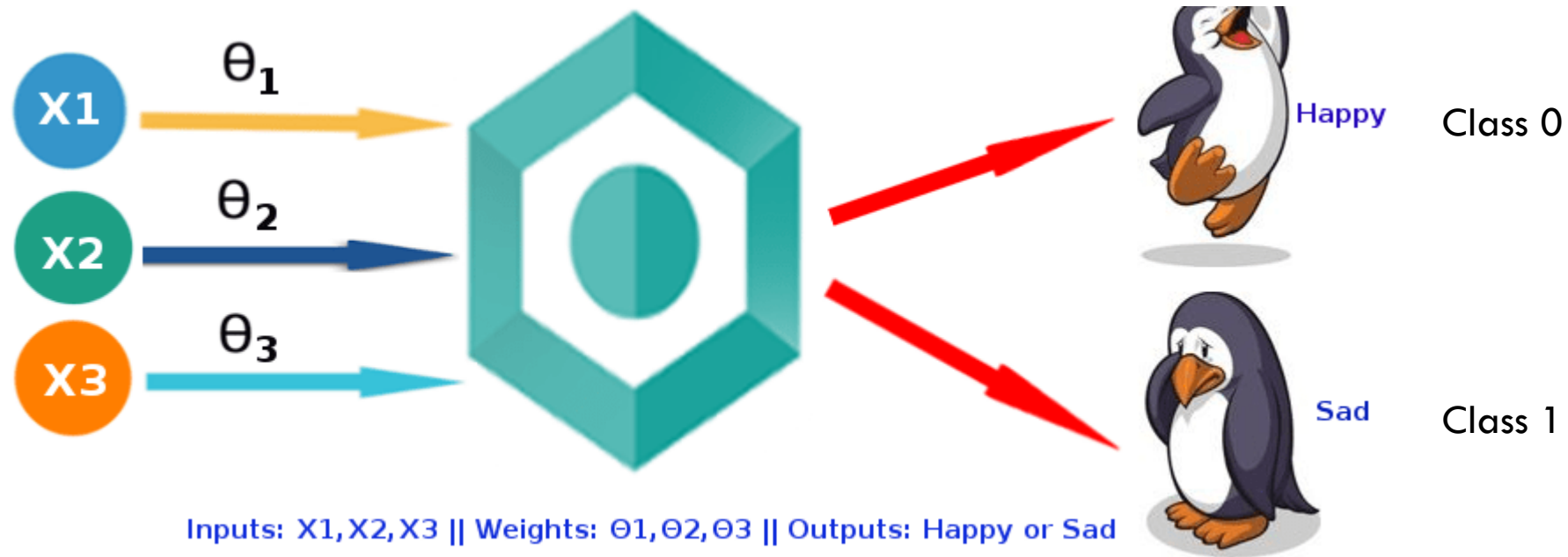


LOGISTIC REGRESSION

Use Case: Classify the emotion of the penguin as whether it is sad or happy

Binary classifier: Class 0 : Happy

Class 1: Sad



LINEAR VS LOGISTIC REGRESSION

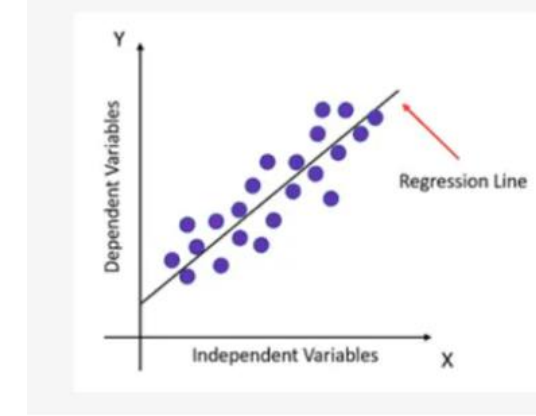
Difference between Linear Regression and Logistic Regression:

Linear Regression	Logistic Regression
Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic Regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.
In Linear regression, we predict the value of continuous variables.	In logistic Regression, we predict the values of categorical variables.
In linear regression, we find the best fit line, by which we can easily predict the output.	In Logistic Regression, we find the S-curve by which we can classify the samples.
Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
The output for Linear Regression must be a continuous value, such as price, age, etc.	The output of Logistic Regression must be a Categorical value such as 0 or 1, Yes or No, etc.
In Linear regression, it is required that relationship between dependent variable and independent variable must be linear.	In Logistic regression, it is not required to have the linear relationship between the dependent and independent variable.
In linear regression, there may be collinearity between the independent variables.	In logistic regression, there should not be collinearity between the independent variable.

In linear regression model given a straight line, y is calculated from independent variable x with coefficients as given by

$$\hat{y} = \beta_0 + \beta_1 X + \epsilon$$

target coefficients input random error



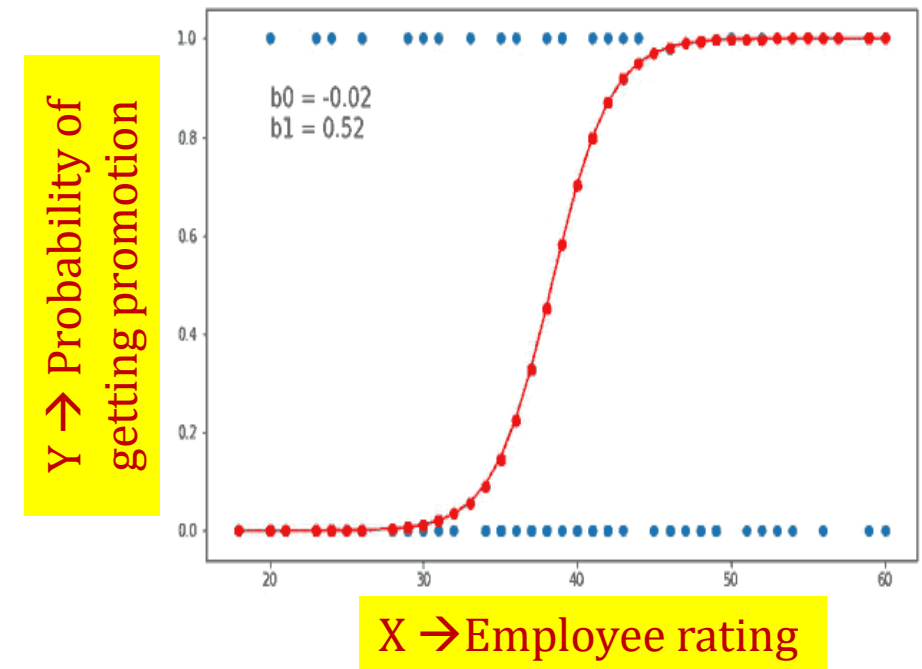
Logistic Regression uses Odds Ratio metric:

An odds ratio (OR) is a statistical measurement that shows how strongly an event is associated with an exposure.

$$\text{odds} = \frac{\hat{p}}{1 - \hat{p}}$$

outputs between 0 and 1

$$g(z) = \frac{1}{1 + e^{-z}} \quad 0 < g(z) < 1$$



The sigmoid function, which squashes the Straight line into a squiggly shaped curve to make the Logistic Regression Model suitable to operate as a classifier (to predict between class 0 and class 1)s

+

↶ ↷

⚙️ ⏪

1



$$y = \left(\frac{1}{1 + E^{(-a)}} \right)$$

×

2



$E = 821$

×

3



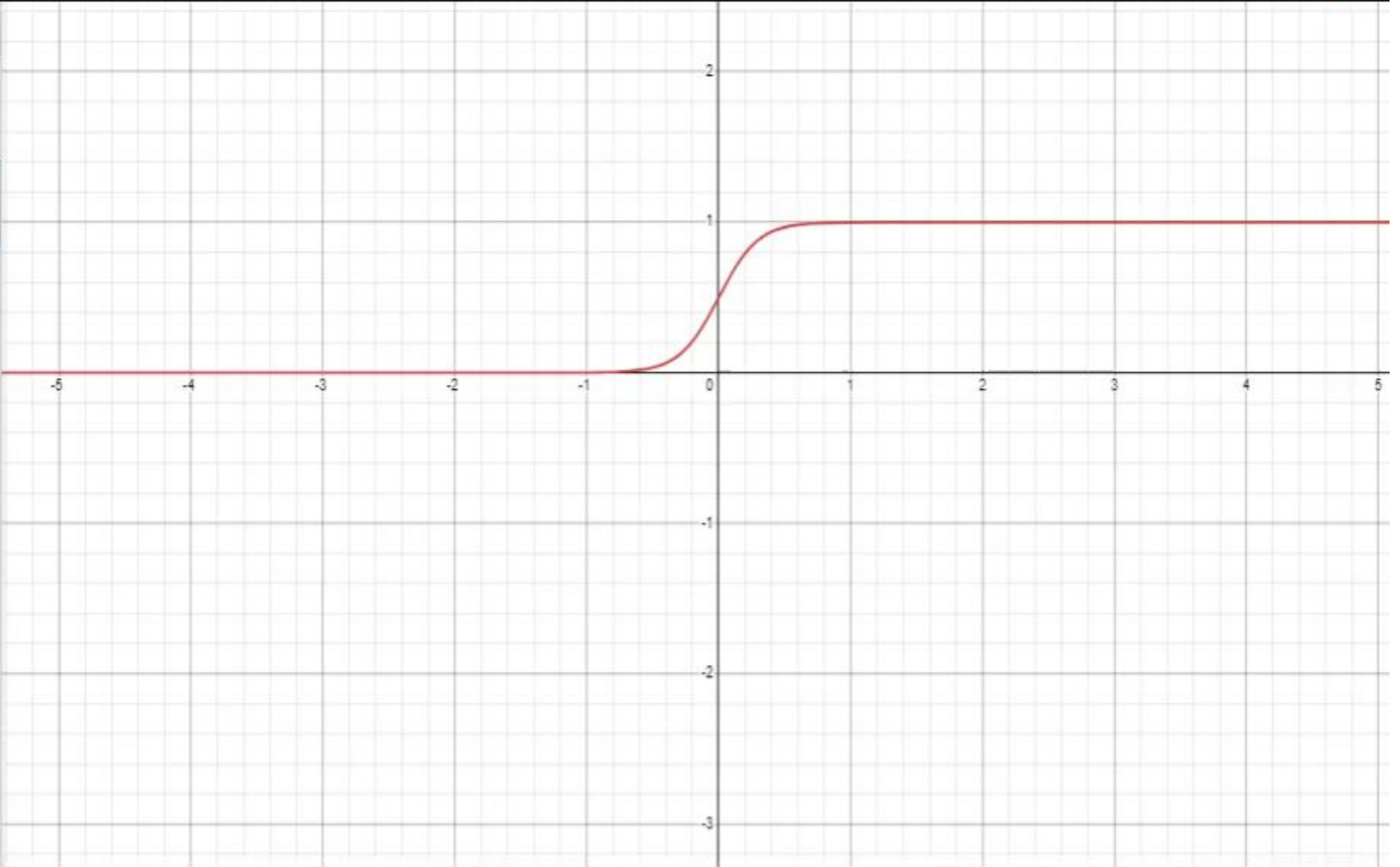
-0.5

1000

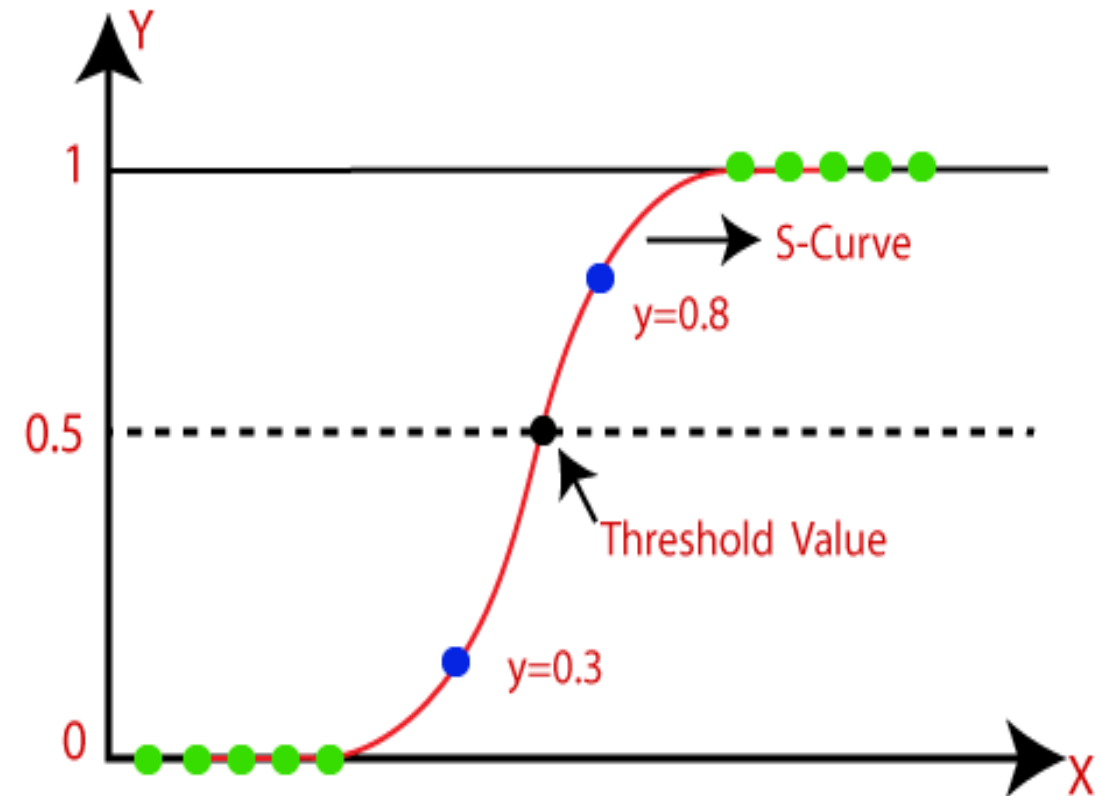
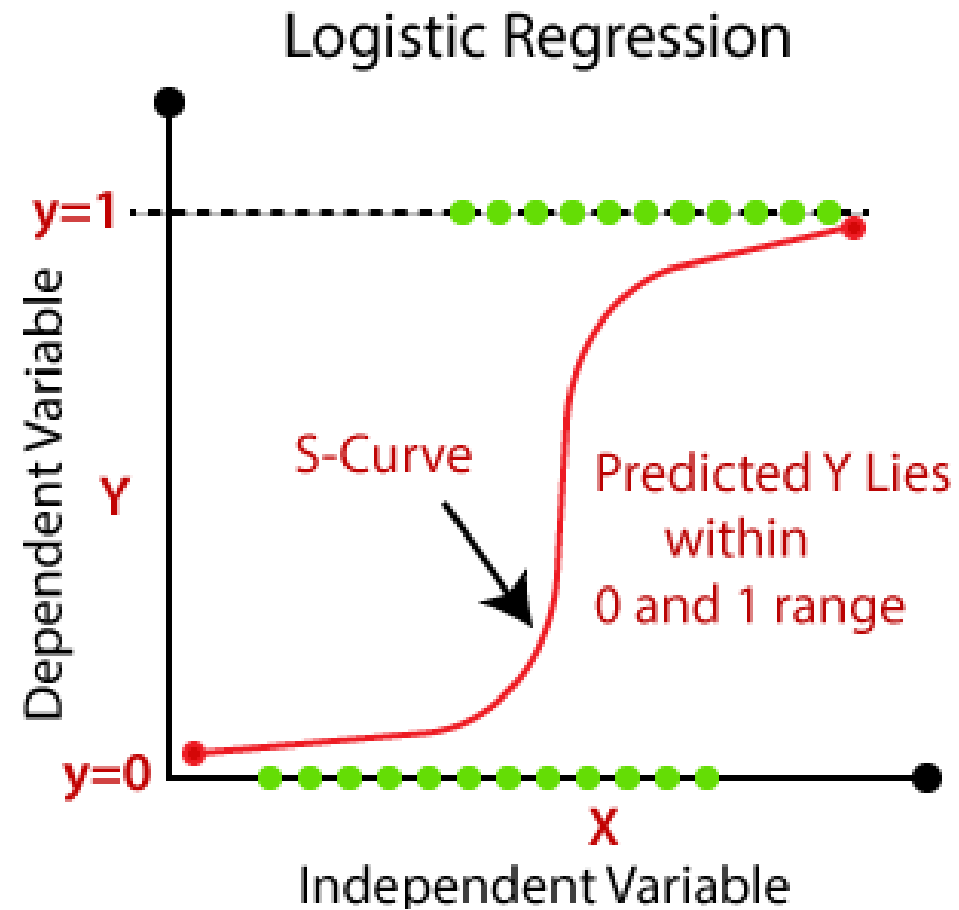
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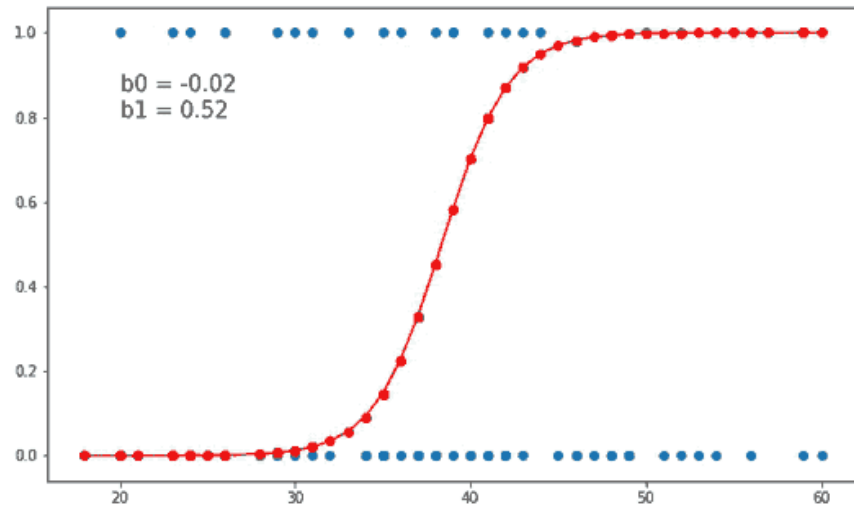
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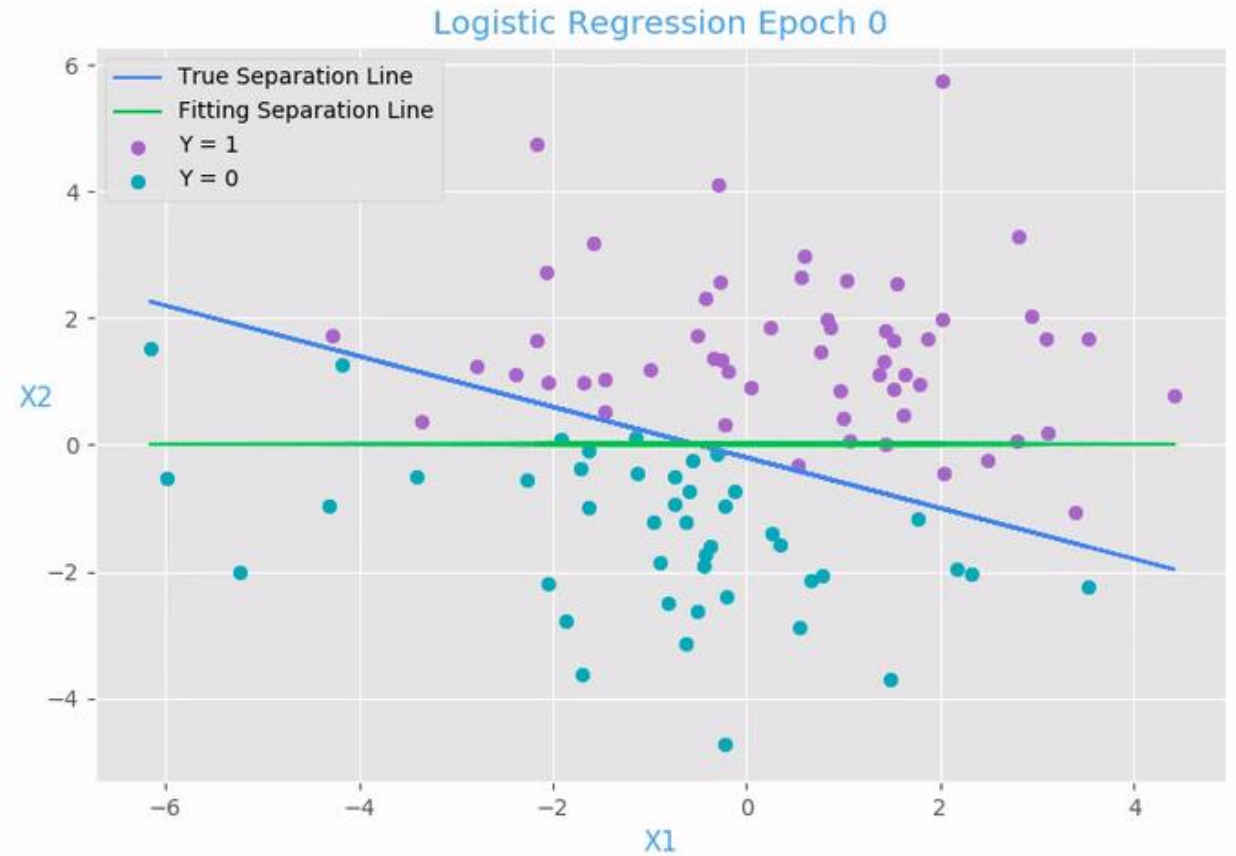
LOGISTIC REGRESSION MODEL



LOGISTIC REGRESSION



$$p(x) = \frac{1}{1 + e^{-(\beta^0 + \beta_1 x)}}$$



LOGISTIC REGRESSION

- ❑ Logistic regression is used to predict the **categorical dependent variable Y** with the help of independent variables (which can be continuous or discrete).
- ❑ The output of Logistic Regression problem can be only between the 0 and 1.
- ❑ Logistic regression can be used where the **probabilities** between two classes is required. Such as whether it will rain today or not, **either 0 or 1, true or false** etc.
- ❑ Logistic regression is based on the concept of **Maximum Likelihood estimation**. According to this estimation, the observed data should be most probable.
- ❑ In logistic regression, we pass the weighted sum of inputs through an activation function that can map values in between 0 and 1. Such activation function is known as **sigmoid function** and the curve obtained is called as sigmoid curve or S-curve. Consider the below image:

Odds ratio

$$odds = \frac{P(\text{occurring})}{P(\text{not occurring})}$$

- Logistic regressions work with **odds ratio** rather than proportions.
- The odds are simply the ratio of the proportions for the two possible outcomes.
- If $\hat{p}$ is the proportion for one outcome, then $1 - \hat{p}$ is the proportion for the second outcome

$$\text{odds} = \frac{\hat{p}}{1 - \hat{p}}$$

The value of probability lies between 0 to 1 and hence the value of odds ratio ranges from 0 to infinity.

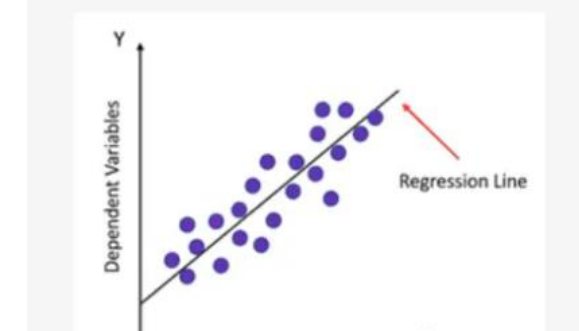
P	Odds
0	0.00
0.1	0.11
0.2	0.25
0.3	0.43
0.4	0.67
0.5	1.00
0.6	1.50
0.7	2.33
0.8	4.00
0.9	9.00
1	#####

$$\begin{aligned} &= 0/1 = 0 \\ &= 0.1/0.9 \\ &= 0.2/0.8 \\ &\vdots \\ &= 1/0 = \infty \end{aligned}$$

In linear regression model given a straight line, y is calculated from independent variable x with coefficients as given by

$$\hat{y} = \beta_0 + \beta_1 X + \epsilon$$

target
coefficients
input
random error



In logistic regression, odds ratio is used and since the value of odds ratio may range from 0 to ∞ , to predict the categorical class labels y the model is subjected to logarithm computation and y is calculated from probability of independent variable x with the coefficients is given as

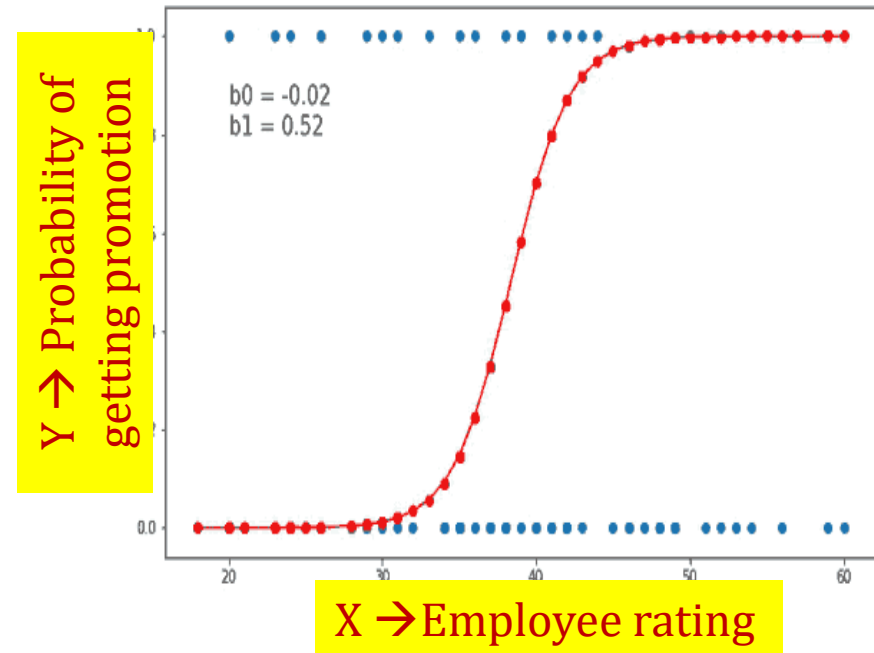
$$p(x) = \frac{1}{1 + e^{-(\beta^0 + \beta_1 x)}}$$

$$\text{Let } z = \beta^0 + \beta_1 x$$

outputs between 0 and 1

$$g(z) = \frac{1}{1 + e^{-z}} \quad 0 < g(z) < 1$$

This is the equation of sigmoid function, which squashes the Straight line into a squiggly shaped curve to make the regression Model to operate as a classifier

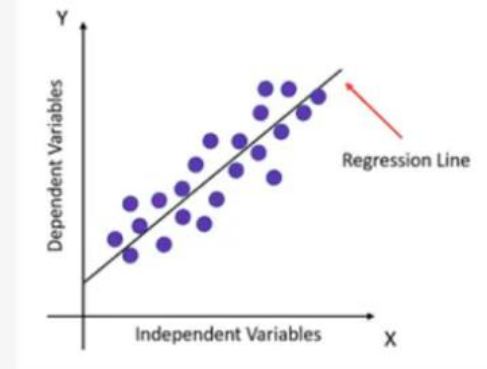


$$p(x) = \frac{1}{1 + e^{-z}}$$

In linear regression model given a straight line, y is calculated from independent variable x with coefficients as given by

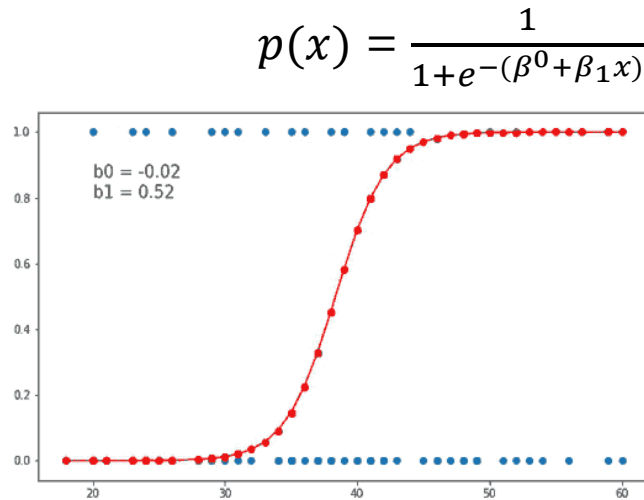
$$\hat{y} = \beta_0 + \beta_1 X + \epsilon$$

target
coefficients
input
random error



Linear Regression Line

In logistic regression, odds ratio is used and since the value of odds ratio may range from 0 to ∞ , to predict the categorical class labels y the model is subjected to logarithm computation and y is calculated from probability of independent variable x with the coefficients is given as



$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

Exponentiating on both sides,

$$e^{\log\left(\frac{p(x)}{1-p(x)}\right)} = e^{\beta_0 + \beta_1 x}$$

$$\frac{p(x)}{1-p(x)} = e^{\beta_0 + \beta_1 x} \rightarrow \text{Eq 1}$$

$$Y = e^{\beta_0 + \beta_1 x} \rightarrow \text{Eq 2}$$

$$Y = \frac{p(x)}{1-p(x)} = e^{\beta_0 + \beta_1 x}$$

$$Y = \frac{p(x)}{1-p(x)}$$

$$(1 - p(x))Y = p(x)$$

$$Y - Y p(x) = p(x)$$

$$Y = p(x) + Yp(x)$$

$$Y = p(x)(1+Y)$$

$$p(x) = \frac{Y}{1+Y} \rightarrow \text{Eq 3}$$

Sub Eq 2 in Eq 3,

$$p(x) = \frac{e^{\beta^0 + \beta_1 x}}{1 + e^{\beta^0 + \beta_1 x}}$$

Now divide numerator and denominator by $e^{\beta^0 + \beta_1 x}$

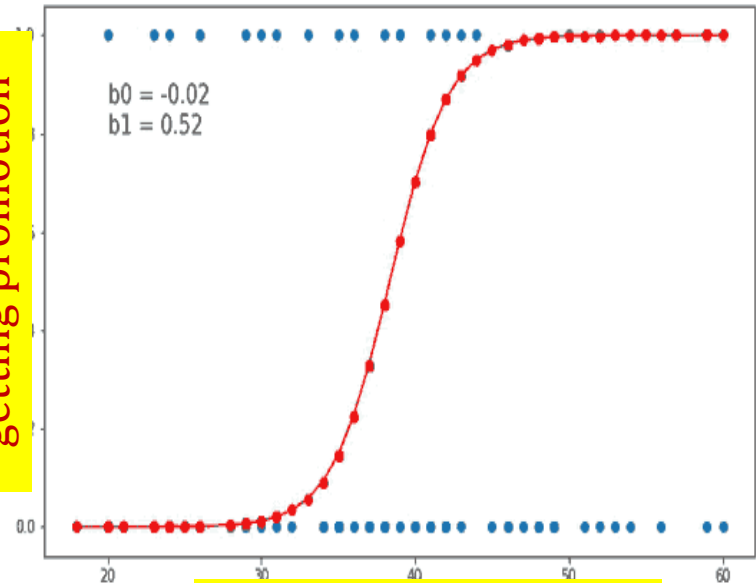
$$p(x) = \frac{e^{\beta^0 + \beta_1 x} / e^{\beta^0 + \beta_1 x}}{1 + e^{\beta^0 + \beta_1 x} / e^{\beta^0 + \beta_1 x}}$$

$$p(x) = \frac{\frac{e^{\beta^0 + \beta_1 x}}{e^{\beta^0 + \beta_1 x}}}{\frac{1}{e^{\beta^0 + \beta_1 x}} + \frac{e^{\beta^0 + \beta_1 x}}{e^{\beta^0 + \beta_1 x}}}$$

$$p(x) = \frac{1}{\frac{1}{e^{\beta^0 + \beta_1 x}} + 1}$$

$$p(x) = \frac{1}{1 + e^{-(\beta^0 + \beta_1 x)}}$$

$Y \rightarrow$ Probability of getting promotion



$X \rightarrow$ Employee rating

This is the equation of sigmoid function, which squashes the Straight line into a squiggly shaped curve to make the regression Model to operate as a classifier

$$\text{Let } z = \beta^0 + \beta_1 x$$

$$p(x) = \frac{1}{1 + e^{-z}}$$

Odds - Example:

- For the customer service data, the proportion of customers who would recommend the service in the sample of customers is $\hat{p} = 0.84$. Find the odds of recommending the service department

$$1 - \hat{p} = 1 - 0.84 = 0.16$$

$$\text{odds} = \frac{\hat{p}}{1 - \hat{p}}$$

$$= \frac{0.84}{0.16}$$

$$= 5.25$$

- ✓ If we round 5.25 to 5 = 5 / 1, we would say that the odds are approximately 5 to 1 that a customer would recommend the service to a friend.
- ✓ In a similar way, we could describe the odds that a customer would not recommend the service as 1 to 5.

LOGISTIC REGRESSION BINARY OUTCOME— LOG LIKELIHOOD

The log likelihood function for logistic regression is typically written as follows:

$$\text{Log Likelihood (LL): } LL(\beta) = \sum_{i=1}^N [y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)]$$

where:

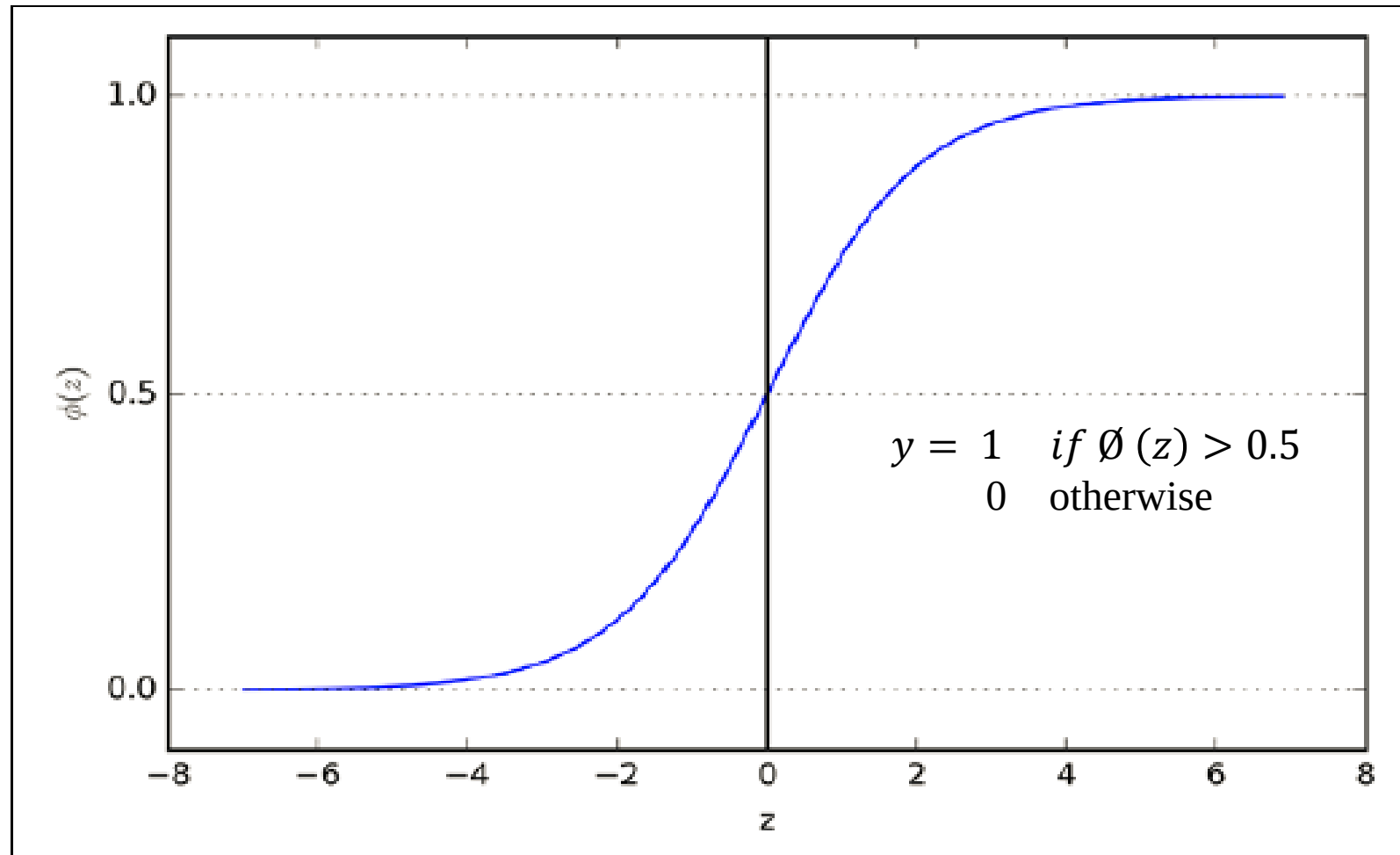
- N is the number of observations.
- y_i is the observed outcome for the i th observation (0 or 1).
- p_i is the predicted probability that the i th observation belongs to the category 1.

The logistic regression model predicts the probability that an observation belongs to a particular category.

The likelihood function in logistic regression describes the probability of observing the given set of outcomes (0s and 1s) based on the logistic regression model.

The log likelihood is the natural logarithm of the likelihood function and is used for mathematical convenience. The goal in logistic regression is to maximize the log likelihood.

S-SHAPED (SIGMOIDAL) CURVE



BINARY OUTCOME (OUTPUT)

Log Likelihood Function (LL): It is the natural logarithm of the likelihood function. The log likelihood is used because it simplifies the mathematical calculations, and the log of a product becomes a sum, making optimization easier.

The log likelihood function for logistic regression is typically written as follows:

Log Likelihood (LL): $LL(\beta) = \sum_{i=1}^N [y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)]$

where:

N is the number of observations.

y_i is the observed outcome for the i th observation (0 or 1).

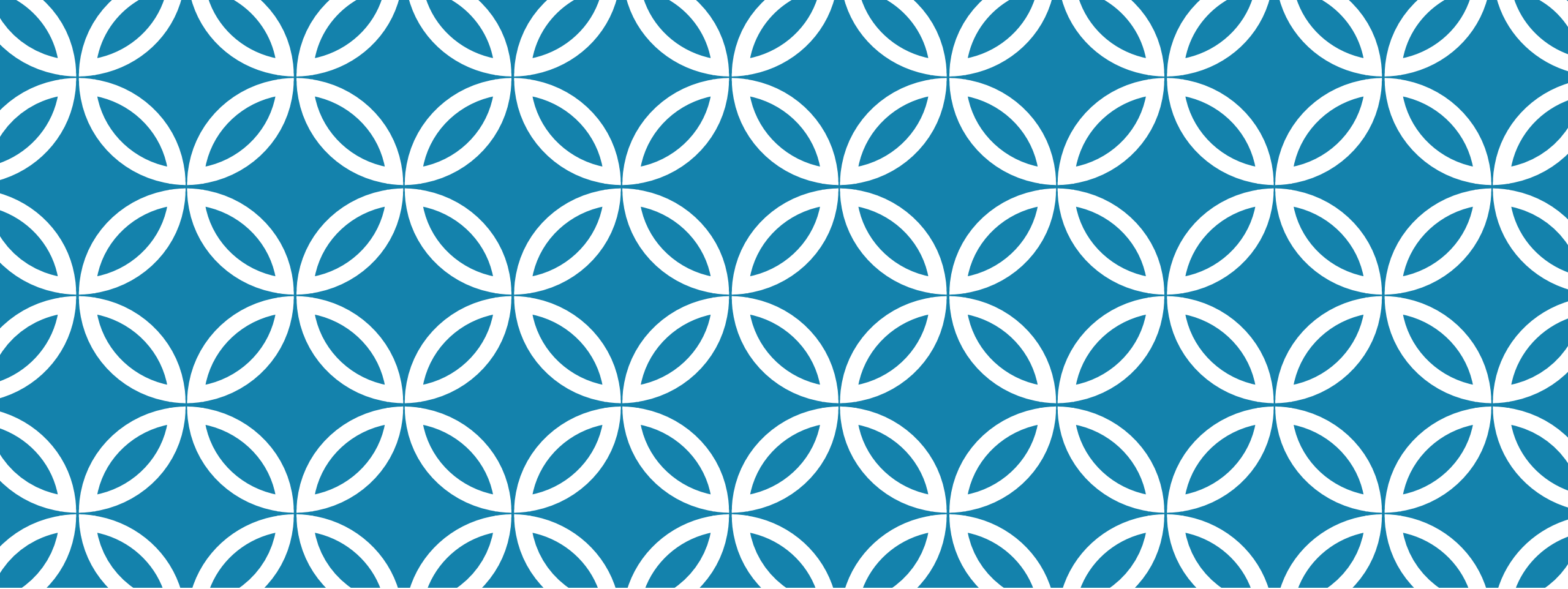
p_i is the predicted probability that the i th observation belongs to the category 1.

Logistic Function (Sigmoid Function):

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1.
- The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

Assumptions for Logistic Regression:

- The dependent variable must be categorical in nature.
- The independent variable should not have multi-collinearity.



NUMERICAL PROBLEM

The dataset of pass/fail in an exam for 5 students is given in the table below. If we use Logistic Regression as the classifier and assume the model suggested by the optimizer will become the following for Odds of passing a course: $\log(\text{Odds}) = -64 + 2 \times \text{hours}$

Instance no	X-Attribute: Hours studied	Result (Pass= 1; Fail=0)
1	29	0
2	15	0
3	33	1
4	28	1
5	39	1

Question Part A) Calculate the probability of Pass for the student who studied 33 hours?

Question Part B) At least how many hours the student should study that makes sure will pass the course with the probability of more than 95%?

Question Part C) How to calculate the loss of model for the student who studied 33 hours?

QUESTION PART A) CALCULATE THE PROBABILITY OF PASS FOR THE STUDENT WHO STUDIED 33 HOURS?

Given that the model is $\log(\text{Odds}) = -64 + 2 \times \text{hours}$

It can be compared with $\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$

As per the dataset given, hours denotes the x attribute, -64 denotes the β_0 and 2 denotes β_1

Therefore, $z = \beta_0 + \beta_1 x = -64 + 2x$

We have to calculate the probability of pass for a student studying 33 hours, so $x = 33$

$$z = -64 + 2x = -64 + 2(33) = 2$$

$$p(x) = \frac{1}{1 + e^{-(-64 + 2(33))}} = \frac{1}{1 + e^{-2}} = 0.881$$

Therefore, the probability of obtaining pass mark while preparing for $x=33$ hours is calculated as 88.1%

QUESTION PART B) AT LEAST HOW MANY HOURS THE STUDENT SHOULD STUDY THAT MAKES SURE WILL PASS THE COURSE WITH THE PROBABILITY OF MORE THAN 95%?

Solution:

$$\text{Given that } p(x) = \frac{1}{1+e^{-z}} = 0.95$$

$$p(x) = \frac{1}{1+e^{-(-64+2(x))}} = 0.95$$

On solving,

$$0.95 (1 + e^{-z}) = 1$$

$$0.95 + 0.95e^{-z} = 1$$

$$e^{-z} = \frac{1-0.95}{0.95} = 0.0526$$

$$\ln(e^{-z}) = \ln(0.0526)$$

$$-z = -2.94$$

Therefore $z = 2.94$

$$z = \beta^0 + \beta_1 x =$$

$$2.94 = -64 + 2(\text{hours})$$

$$64 + 2.94 = 2(\text{hours})$$

$$\text{hours} = \frac{66.94}{2} = 33.47$$

hours = 33.47 hours

Therefore, the student has to spend atleast 33.47 hours to get pass with a probability of 95%.

QUESTION PART C) HOW TO CALCULATE THE LOSS OF MODEL FOR THE STUDENT WHO STUDIED 33 HOURS?

Instance no	X-Attribute: Hours studied	Result (Pass= 1; Fail=0)
1	29	0
2	15	0
3	33	1
4	28	1
5	39	1

x	$p(x) = \frac{1}{1+e^{-(\beta^0+\beta_1x)}}$ Probability	$Logloss = -\frac{1}{N} \sum_{i=1}^N (y_i \log(p_i) + (1 - y_i) \log(1 - p_i))$
33	$p(33) = \frac{1}{1+e^{-(-64+2(33))}} = \frac{1}{1+e^{-(2)}} = 0.88$	$Loss = (y_i \log(p_i) + (1 - y_i) \log(1 - p_i)) = 1(\ln(0.88)) + (1-1)(\ln(1-0.88))$ $Loss = -0.128$

Instance no	X-Attribute: Hours studied	Result (Pass= 1; Fail=0)
1	29	0
2	15	0
3	33	1
4	28	1
5	39	1

QUESTION PART D) WHAT IS THE TOTAL LOSS OF THE MODEL GIVEN IN EQUATION BELOW?

$$\text{Logloss} = -\frac{1}{N} \sum_{i=1}^N (y_i \log(p_i) + (1 - y_i) \log(1 - p_i))$$

x	$p(x) = \frac{1}{1+e^{-(\beta_0+\beta_1x)}}$ Probability	$(y_i \log(p_i) + (1 - y_i) \log(1 - p_i))$	Log loss
29	$p(29) = \frac{1}{1+e^{-(-64+2(29))}} = \frac{1}{1+e^{-(-6)}} = 2.47e^{-3}$	$= (0 \log(2.47e^{-3}) + (1 - 0) \log(1 - 2.47e^{-3}))$	$-2.4e^{-3}$
15	$p(15) = \frac{1}{1+e^{-(-64+2(15))}} = \frac{1}{1+e^{-(-34)}} = 1.714e^{-15} = 0$	$= (0 \log(1.714e^{-15}) + (1 - 0) \log(1 - 1.714e^{-15}))$	0
33	$p(33) = \frac{1}{1+e^{-(-64+2(33))}} = \frac{1}{1+e^{-(2)}} = 0.88$	$= (1 \log(0.88) + (1 - 1) \log(1 - 0.88))$	-0.128
28	$p(28) = \frac{1}{1+e^{-(-64+2(28))}} = \frac{1}{1+e^{-(-8)}} = 3.35e^{-4}$	$= (1 \log(3.35e^{-4}) + (1 - 1) \log(1 - 3.35e^{-4}))$	-8.016
39	$p(39) = \frac{1}{1+e^{-(-64+2(39))}} = \frac{1}{1+e^{-(14)}} = 0.999$	$= (1 \log(0.999) + (1 - 1) \log(1 - 0.999))$	-0.001
Total log loss = $-\frac{1}{N} \sum_{i=1}^N (y_i \log(p_i) + (1 - y_i) \log(1 - p_i))$			$-(1/5)(-2.4E-3 + 0 - 0.128 - 8.016 - 0.001) = 1.6296$

The total loss of the model is 1.6296